

★ WEEK 12 · PART 2 · PRACTICE

The Term Structure, Worked by Hand

Price a Bond, Then Build the Whole Curve

From the affine Vasicek formula to yields and shapes

Part 2's term structure, with the arithmetic shown. We price a zero-coupon bond with the affine Vasicek formula and turn it into a yield, then build the whole curve from today's short rate - and watch it flip from upward to inverted.

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PART 2 · APPLICATIONS

One Short Rate, a Whole Curve

Case A: price a 5-year zero with the affine Vasicek formula, then convert it to a yield. Case B: compute yields across maturities for a low and a high short rate, seeing the curve tilt from upward to inverted. Then a Claude Code lab.

What Part 2 Told Us

Now we price and build - with numbers

1

A bond price is an expectation

Discount the future at the random short rate and average over its paths. Vasicek makes that average a clean formula.

2

The Vasicek price is affine

Price equals A times e -to-the-minus- B -times- r . Take a log and divide by maturity, and you have the yield.

3

Today's short rate sets the shape

One number tilts the whole curve - upward when rates are low, inverted when they're high. We compute both.

Everything here is a calculator exercise: one bond price, one yield, two curve shapes.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the affine Vasicek formula.
- Compute a price, a yield, or a curve.
- Then read the shape it produces.

What you should watch

- How the affine formula turns r into a price.
- How price becomes yield in one step.
- How the short rate flips the curve's slope.

Price a Vasicek Bond

From the short rate to a yield

Under Vasicek, a zero-coupon bond has an affine price: a maturity-dependent factor A times an exponential in today's short rate, with a coefficient B . We compute B and A for a five-year bond, price it at a 4% short rate, and turn that price into a yield.

The setup

- Vasicek: $\kappa = 0.30$, $\theta = 5\%$, $\sigma = 2\%$.
- Today's short rate is 4%.
- Maturity is five years.

What we'll do by hand

- Compute $B(5)$ and $A(5)$.
- Price the bond as A times e -to-the-minus- $B \cdot r$.
- Convert the price to a yield.

The Affine Bond Price

Price, linear in the short rate's exponent

$$P(t, T) = A(t, T) e^{-B(t, T) r_t}, \quad B(t, T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa} \quad (\text{price is affine in the short rate})$$

The B factor

- It measures sensitivity to the short rate.
- It grows with maturity, then levels off.
- It's the duration-like term.

The A factor

- It captures the drift and volatility terms.
- It depends only on maturity, not on r .
- Together they give the full price.

Compute B and A

The two affine factors

$$B(5) = \frac{1 - e^{-0.30 \times 5}}{0.30} = \frac{1 - 0.223}{0.30} = \mathbf{2.590}$$
$$\ln A(5) = -0.1174 \Rightarrow A(5) = \mathbf{0.889}$$

B(5) is 2.59 - the bond's sensitivity to the short rate over five years - and A(5) works out to 0.889. Both depend only on the parameters and the maturity.

The Price and the Yield

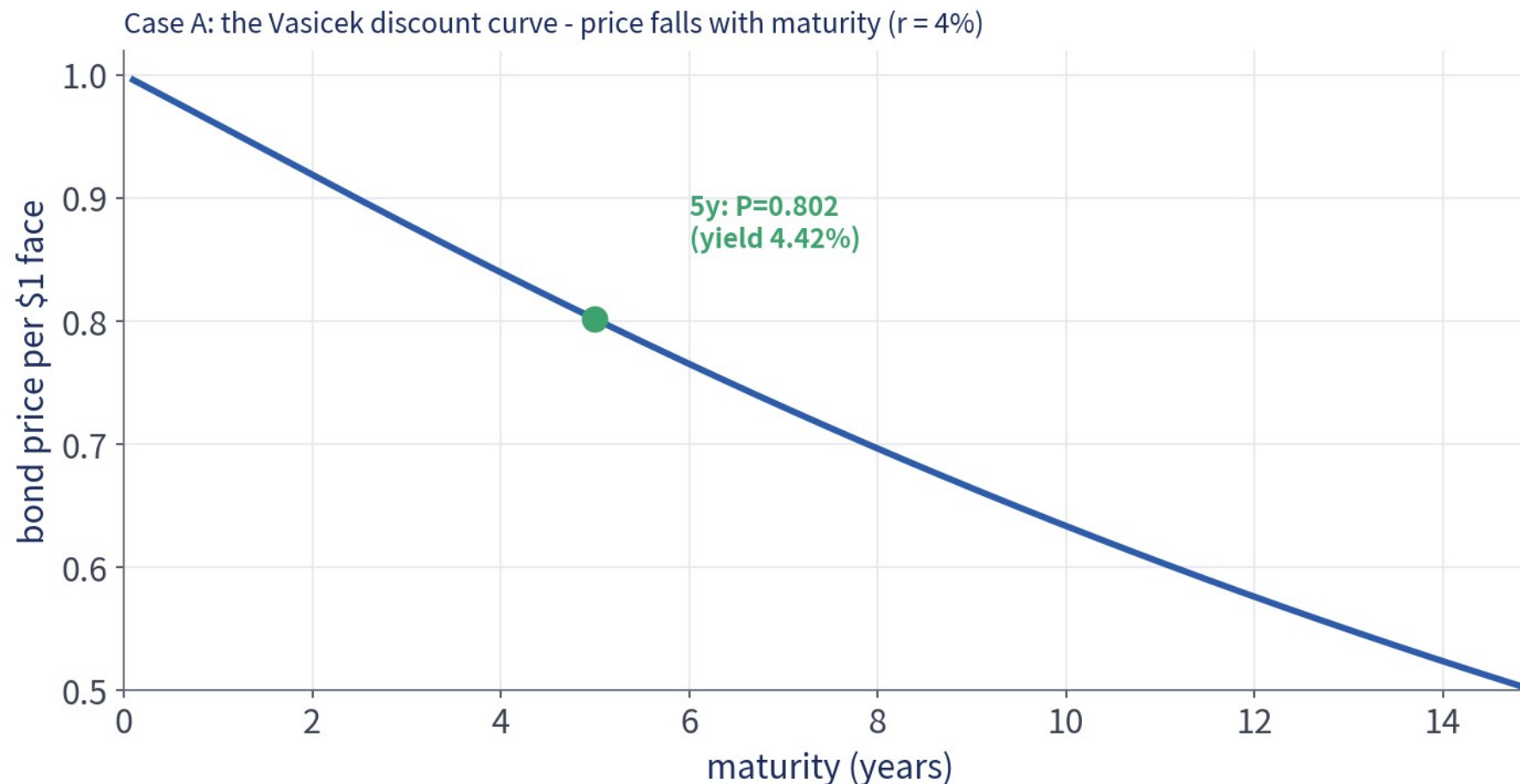
One step from price to curve

$$P(5) = A e^{-Br} = 0.889 e^{-2.590(0.04)} = \mathbf{0.8017}$$
$$y(5) = -\frac{\ln P(5)}{5} = \frac{0.2210}{5} = \mathbf{4.42\%}$$

Plugging in the 4% short rate gives a price of 0.8017 per dollar of face. Taking minus the log over the maturity turns that into a 4.42% yield.

The Discount Curve

Pricing bonds of every maturity at the 4% short rate traces the discount curve: the price falls smoothly as maturity lengthens, because more discounting is applied. The five-year point sits at 0.802 - the bond we priced, yielding 4.42%.



Why Affine Is Powerful

A whole curve from one formula

What affine buys you

- A closed-form price - no simulation needed.
- Every maturity from the same two factors.
- Yields that are linear in the short rate.

Why it matters

- Fast to compute and to calibrate.
- Transparent - you see what drives the price.
- The backbone of tractable rate models.

Build the Curve, Read the Shape

Today's short rate does the tilting

Because the yield is linear in the short rate, one number sets the whole curve. When the short rate sits below the long-run level, the curve slopes up; when it sits above, the curve inverts. We compute yields at four maturities for a low and a high short rate and watch the shape flip

The two scenarios

- Low short rate: 2%, below the long run.
- High short rate: 8%, above the long run.
- Both converge to about 4.78%.

What we'll do by hand

- Compute yields at 1, 3, 5, and 10 years.
- Do it for both short rates.
- Read the slope of each curve.

The Yield, Linear in r

How the curve is built

$$y(T) = \frac{B(T)}{T} r - \frac{\ln A(T)}{T} \quad (\text{yield is linear in today's short rate } r)$$

The short end

- At short maturities, B over T is near one.
- So the yield tracks the short rate closely.
- The near end anchors on today's r .

The long end

- As maturity grows, the yield forgets r .
- It converges to the long-run level.
- About 4.78% for these parameters.

Low Rate: an Upward Curve

Short rate 2%, below the long run

maturity	1y	3y	5y	10y	shape
yield ($r=2\%$)	2.40%	2.99%	3.38%	3.93%	UPWARD

Starting below the long-run level, the yield climbs with maturity toward 4.78% - the normal, upward-sloping curve seen in ordinary times.

Why it rises

- The short rate is pulled up toward theta.
- Longer bonds average in that higher future.
- So longer yields exceed the short rate.

The reading

- An upward curve is the common case.
- It signals expected mean reversion upward.
- No stress in the rate outlook.

High Rate: an Inverted Curve

Short rate 8%, above the long run

maturity	1y	3y	5y	10y	shape
yield ($r=8\%$)	7.59%	6.95%	6.49%	5.83%	INVERTED

Starting above the long-run level, the yield falls with maturity - the inverted curve, historically a recession signal. produced by nothing more than a high short rate.

Why it falls

- The short rate is pulled down toward theta.
- Longer bonds average in that lower future.
- So longer yields fall below the short rate.

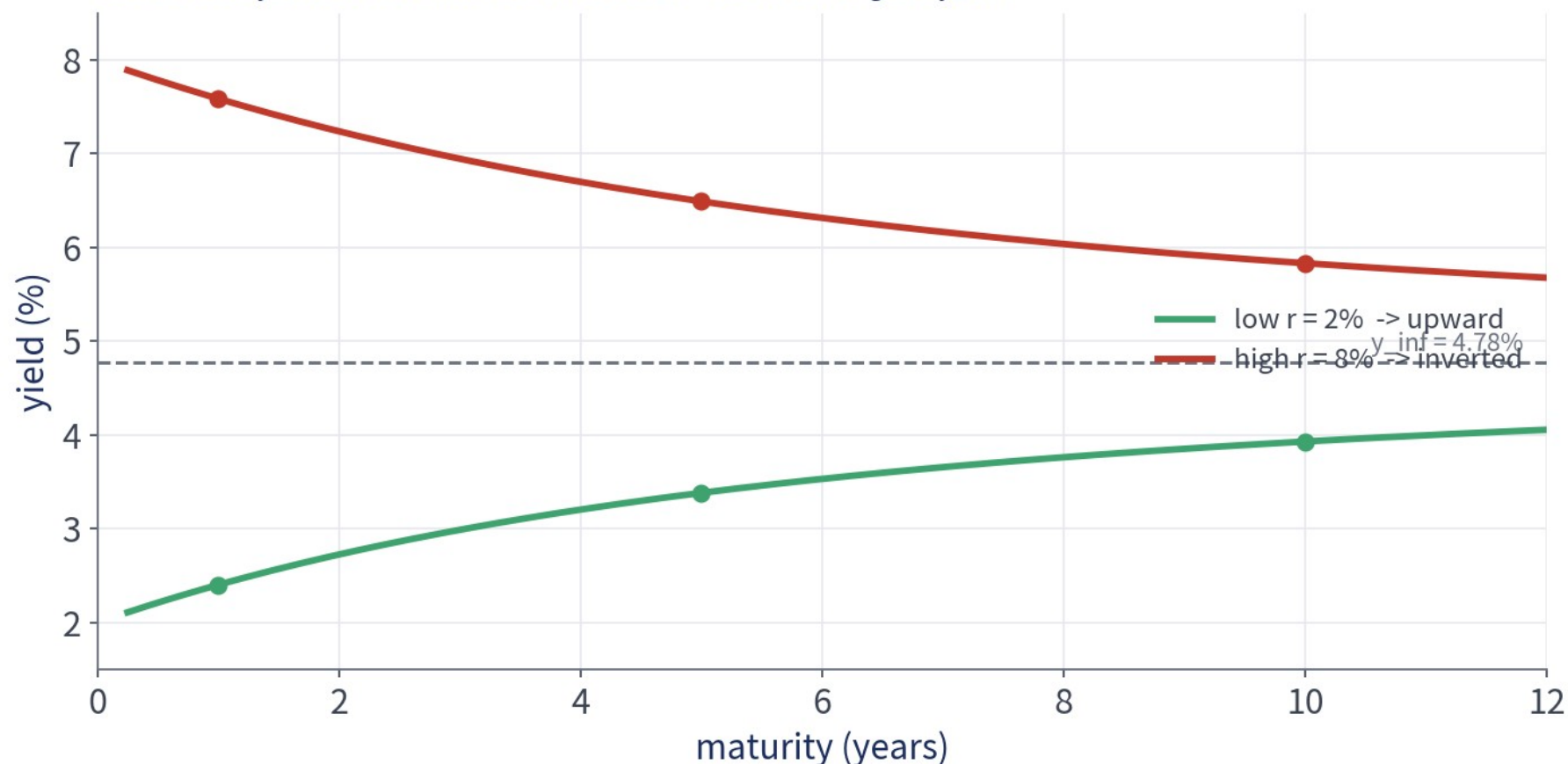
The reading

- An inverted curve flags expected easing.
- The same formula, a different short rate.
- Shape is set entirely by where r starts.

The Short Rate Tilts the Curve

From a low short rate the curve slopes up; from a high one it inverts. Both fan out from today's rate at the short end and converge to the same long-run level of 4.78% at the long end. The entire shape is decided by where the short rate starts

Case B: today's short rate tilts the whole curve - both converge to y_{inf}



The Curve Tells a Story

Shape as a signal

What the shape encodes

- Upward: rates expected to rise toward theta.
- Inverted: rates expected to fall toward theta.
- Humped: a mix in between.

The caveat

- This is the model's mechanism, not a forecast.
- Real curves also carry risk premia.
- But the intuition - r sets the tilt - holds.

One Rate, One Formula, One Curve

The affine engine

The whole term structure comes from one short rate through one affine formula. Price is A times $e^{-B \cdot r}$; yield is that, logged and scaled by maturity. Feed in today's rate and every maturity's yield falls out - and its position relative to the long-run level decides whether the curve rises, inverts, or humps.

Pricing (Case A)

- Affine price from B and A .
- 5-year bond at 0.802, yielding 4.42%.
- Closed form, no simulation.

The curve (Case B)

- Yields linear in the short rate.
- Low r upward, high r inverted.
- All converging to 4.78%.

Reproduce the Price and the Curve

Direct Claude Code to check your arithmetic

GOAL Verify the affine bond price and yield, then plot curves for low and high rates.

[INPUT] Prompt · Claude Code

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Write and run part2_practice.py that, for Vasicek
kappa=0.30, theta=0.05, sigma=0.02:
CASE A - price a bond:
    * compute B(5), A(5), the 5y price at r=0.04,
      and the yield -ln(P)/5.
CASE B - the curve:
    * print yields at 1,3,5,10y for r=0.02 and r=0.08;
    * label each curve upward or inverted;
    * plot both and mark the long-run yield.
Match every printed number to the slides.
```

Check against the slides

- $B(5)=2.59$, $A(5)=0.889$.
- $P(5)=0.802$, yield 4.42%.
- low r: 2.40 -> 3.93% (up).
- high r: 7.59 -> 5.83% (inverted).

Deliverable: part2_practice.py whose printed numbers match the bond price and the two curves.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What makes the Vasicek price affine?

Price = A times e-to-the-minus-B-times-r

Q2 How do you turn a bond price into a yield?

Minus the log of the price over the maturity

Q3 What decides the curve's shape?

Today's short rate, relative to the long run

All three come from the affine bond price and the yield built from it.

Three Pitfalls to Avoid

Where term structure trips people

Common mistakes

- "The whole curve moves with the short rate equally."
- "An inverted curve is a direct recession forecast."
- "The long yield equals the long-run mean theta."

The reality

- The short end moves most; the long end barely budges.
- It's the model's mechanism - real curves add risk premia.
- The long yield is theta minus a volatility term (4.78%).

Two Calculations to Remember

Part 2 applications in one screen

1

Price a bond with the affine formula

For a 5-year Vasicek bond, $B(5) = 2.59$ and $A(5) = 0.889$; at a 4% short rate the price is 0.8017 per dollar of face, and minus its log over five years gives a 4.42% yield. A closed form, no simulation.

2

Build the curve from the short rate

Because yields are linear in r , a low 2% rate produced an upward curve (2.40% to 3.93%) and a high 8% rate an inverted one (7.59% to 5.83%), both converging to the 4.78% long-run level. The short rate tilts everything.

3

One rate, one formula, one curve

The affine structure turns a single short rate into a price and a yield for every maturity. Where that rate sits relative to the long-run level decides whether the curve rises, inverts, or humps.

Next — Part 3 practice: discretizing and calibrating these models to real market data.